

TARGET 14.1

QUESTION 1 • TARGET AIM

SPEAKER 1

Target 14.1: Eliminate Marine Pollution

Halting Land-Based Agricultural Runoff and Plastic Influx by 2025

ORIGIN POINT

80% Land-Based

Massive municipal waste, toxic chemicals, and fertilizers flowing down river basins.

GLOBAL MANDATE

Target 2025

Official UN deadline to prevent and significantly reduce marine pollution of all kinds.

PRIMARY VECTORS

N, P & Plastics

Excess reactive nitrogen, phosphorus, and macro to microplastics choking marine life.

TARGET 14.1

QUESTION 2 • OFFICIAL INDICATORS

SPEAKER 1

Indicator 14.1.1: Dual Monitoring Pathways

Tracking Invisible Fertilizer Nutrients and Visible Solid Plastic Accumulation

SUB-INDICATOR 14.1.1(A)

ICEP Index

Index of Coastal Eutrophication Potential monitoring nitrogen and phosphorus chemical runoff.

SUB-INDICATOR 14.1.1(B)

Plastic Density

Mass and particle count of macro and micro-plastics per km² of seawater and shoreline.

ECOLOGICAL IMPACT

500+ Dead Zones

Hypoxic ocean dead zones worldwide stripped of dissolved oxygen by nutrient-fed algal blooms.

TARGET 14.1

QUESTION 3 • MEASUREMENT METHODS

SPEAKER 2

Measuring Pollution: Orbit to Shoreline

Satellite Remote Sensing Combined with Standardized Coastal Transects

SATELLITE SURVEILLANCE

MODIS & Sentinel

ESA Sentinel-3 & NASA orbiters scan ocean optical anomalies to detect Chlorophyll-a spikes.

COASTAL TRANSECTS

100m Shoreline

Standardized beach surveys categorizing, weighing, and geolocating macroplastic debris.

SURFACE TRAWLING

Manta Nets (333 μ m)

Fine mesh surface trawls capturing microplastics for FTIR and Raman laboratory spectroscopy.

TARGET 14.1

QUESTION 4 • INSTITUTIONS & GOVERNANCE

SPEAKER 2

Governance 14.1: Treaties & Custodianship

Global Coordination Under UNEP and the Binding Global Plastics Treaty

GLOBAL CUSTODIAN

UNEP

United Nations Environment Programme
standardizes national reporting methodologies.

LEGALLY BINDING PACT

INC Plastics Treaty

Historic Intergovernmental Negotiating
Committee treaty tackling the entire plastic
lifecycle.

MARITIME LAW

MARPOL Annex V

IMO convention strictly banning commercial
vessel dumping of plastics and garbage at sea.

TARGET 14.2

QUESTION 1 • TARGET AIM

SPEAKER 3

Target 14.2: Protect & Restore Ecosystems

Shifting from Sectoral Silos to Holistic Ecosystem-Based Management (EBM)

CORE STRATEGY

EBM Approach

Ecosystem-Based Management: regulating human pressures across whole integrated biomes.

CRITICAL HABITATS

Reefs & Mangroves

Active conservation of coastal storm buffers, blue carbon sinks, and larval fish nurseries.

TIMELINE MANDATE

Urgent Catch-up

Target 14.2 held an early 2020 deadline; national implementations are in aggressive acceleration.

TARGET 14.2

QUESTION 2 • OFFICIAL INDICATORS

SPEAKER 3

Indicator 14.2.1: Management on Paper

National Policy Adoption vs. On-the-Ground Ecological Reality

UN INDICATOR 14.2.1

Policy Adoption

Number of countries adopting ecosystem-based approaches to manage marine areas.

SPATIAL INSTRUMENTS

MSP & ICZM

Marine Spatial Planning and Integrated Coastal Zone Management national legal decrees.

CRITICAL LIMITATION

Paper Parks

Indicator measures enacted paperwork—it does not prove coral recovery or enforced patrols.

TARGET 14.2

QUESTION 3 • MEASUREMENT METHODS

SPEAKER 4

Measuring Ecology: Sensors, eDNA & Sonar

Direct Biological Auditing Beyond Bureaucratic Self-Reporting

REEF HEALTH NETWORK

GCRMN Censuses

Global Coral Reef Monitoring Network standardizing live hard coral cover and bleaching rates.

GENOMIC SAMPLING

eDNA Sequencing

Filtering seawater DNA to profile all fish species present without destructive netting.

ACOUSTIC BATHYMETRY

Multibeam Sonar

3D acoustic mapping of benthic rugosity, structural reef complexity, and seagrass meadows.

TARGET 14.2

QUESTION 4 • INSTITUTIONS & GOVERNANCE

SPEAKER 4

Governance 14.2: 30x30 & The High Seas

The Kunming-Montreal Framework and the Historic BBNJ High Seas Treaty

GLOBAL BIODIVERSITY PACT

30x30 Target

Kunming-Montreal GBF Target 3: legally designate 30% of global oceans as MPAs by 2030.

INTERNATIONAL WATERS

BBNJ Treaty

Biodiversity Beyond National Jurisdiction: first UN instrument to enforce high-seas protected zones.

LEAD AGENCIES

UNEP / CBD

UNEP harmonizes compliance with the Convention on Biological Diversity Secretariat.

TARGET 14.3

QUESTION 1 • TARGET AIM

SPEAKER 5

Target 14.3: Minimize Ocean Acidification

Confronting the Severe Chemical Cost of Planetary Carbon Absorption

PLANETARY SINK

25% CO₂ Absorbed

Oceans absorb a quarter of anthropogenic carbon emissions, creating carbonic acid (H₂CO₃).

ACIDITY SURGE

+30% Acidity

A 0.1 unit drop in global surface seawater pH represents a 30% surge in hydrogen ion (H⁺) concentration.

CALCIFIER CRISIS

Corrosive Waters

Depletion of carbonate ions (CO₃²⁻) dissolves shells of pteropods, shellfish, and coral skeletons.

TARGET 14.3

QUESTION 2 • OFFICIAL INDICATORS

SPEAKER 5

Indicator 14.3.1: Precision Marine Chemistry

Average Acidity (pH) Measured at Representative Sampling Stations

PRIMARY CHEMICAL METRIC

Total pH Scale (pHT)

Seawater pH measured continuously to high scientific precision of ± 0.002 units.

CARBON SYSTEM PARTNERS

DIC & Alkalinity

Reporting Total Dissolved Inorganic Carbon and Total Alkalinity alongside pH for full thermodynamic equations.

SATURATION HORIZON

Ω Aragonite

Tracking shoaling aragonite saturation depth where deep ocean currents become corrosive to bone.

TARGET 14.3

QUESTION 3 • MEASUREMENT METHODS

SPEAKER 6

Measuring Chemistry: Autonomous Robots & Arrays

Autonomous Biogeochemical Profiling Across the Global Water Column

ROBOTIC PROFILERS

BGC-Argo Floats

Autonomous robotic floats equipped with ISFET pH sensors diving continuously to 2,000 meters.

SHIPBOARD PURITY

m-Cresol Purple

Purified dye spectrophotometry aboard research vessels calibrated against Certified Reference Materials.

GLOBAL NETWORK

GOA-ON Network

Global Ocean Acidification Observing Network coordinating thousands of buoys and ships across 100+ nations.

TARGET 14.3

QUESTION 4 • INSTITUTIONS & GOVERNANCE

SPEAKER 6

Governance 14.3: UNESCO–IOC & Global Action

Bridging the North–South Data Divide and Uniting All Three Targets

CUSTODIAN LEAD

UNESCO–IOC

Intergovernmental Oceanographic Commission aggregates and publishes the global SDG 14.3.1 data portal.

CAPACITY EQUITY

OA in a Box

Deploying standardized low-cost chemical laboratory kits to Small Island Developing States (SIDS).

ROOT REMEDY

UNFCCC Paris Pact

Ocean acidification has no local engineering cure —its only solution is reducing global fossil emissions.